

Project Title: "Number guessing game"

Student 1 Details:

Name: Fariha Tabassum

ID: 242300

AUW email: auw242300@auw.edu.bd

Student 2 Details:

Name: Shimu Jadab

ID: 242056

AUW email: auw242056@auw.edu.bd

Student 3 Details:

Name: Salma Ayesha

ID: 232178

AUW email: auw232178@auw.edu.bd

Submission date: 13th august, 2026

Project Description: The **Number Guessing Game** is a simple console-based C programming project designed to allow multiple players to play a number guessing game. In this game, the target number is fixed at **50**. Each player is given a maximum of **five attempts** to guess the correct number.

The program provides a menu with four options:

1. Play Game
2. Display Players
3. Search Player
4. Exit

Introduction: In the game, the computer has a fixed target number, which is **50**. The player is asked to guess the number.

The player has a maximum of five attempts. After every incorrect guess, the program provides a hint by displaying either "**Too Low**" or "**Too High**".

The scoring system starts at **50 points**. Each incorrect guess decreases the score by **10 points**. If the player guesses the number correctly, the game ends immediately and the player's current score is recorded. If the player fails to guess the number after five attempts, the score becomes zero and the program displays "**Game Over**."

The program also maintains a list of players and their scores. This allows users to display all registered players and search for a specific player's score.

Objectives:

1. To practice C programming
2. To use functions
3. To understand arrays
4. To practice pointers
5. To practice conditional statements
6. To implement searching
7. To create a user-friendly program

Methodology/Implementation:

1. Program Planning
2. Storing Player Information
3. Implementing the Game
4. Using Pointers

5. Displaying Player Information
6. Searching for a Player
7. Menu and Control Flow
8. Testing the Program
9. Input and Output
10. Final Implementation

Output:

1. Play Game
2. Display Players
3. Search Player
4. Exit

Enter choice: 1

Enter player name: Alice

Enter guess: 50

Correct!

Enter choice: 1

Enter player name: Bob

Enter guess: 30

Too Low

Enter guess: 70

Too High

Enter guess: 50

Correct!

Player List

Alice 50

Bob 30

Enter player name: Bob

Player Found

Name: Bob

Score: 30

Enter player name: David

Player Not Found

Challenges and limitations:

1. **Managing Multiple Players**
2. Using Pointers to Update Scores
3. Handling the Number of Attempts
4. Correct score calculation
5. Searching Player Names

6. Fixed Target Number

7. No Permanent Data Storage

Limitations: Although the project successfully demonstrates several fundamental C programming concepts, it is still a basic console-based application. It does not currently include random target numbers, permanent data storage, advanced input validation, difficulty levels, or a leaderboard.

Member's contribution:

Member 1:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define MAX_PLAYERS 50
```

```
void playGame(int *score);
```

```
void displayPlayers(char names[][30], int scores[], int count);
```

```
void searchPlayer(char names[][30], int scores[], int count, char  
search[]);
```

```
int main()
```

```
{
```

```
    char players[MAX_PLAYERS][30];
```

```
int scores[MAX_PLAYERS];

int count = 0;

int choice;

char search[30];


while(1)
{
    printf("\n1. Play Game\n2. Display Players\n3. Search
Player\n4. Exit\n");

    printf("Enter choice: ");

    scanf("%d",&choice);


    switch(choice)
    {

        case 1:

            printf("Enter player name: ");

            scanf("%s", players[count]);


            playGame(&scores[count]);

            count++;
```

```
break;
```

```
case 2:
```

```
    displayPlayers(players, scores, count);
```

```
    break;
```

```
case 3:
```

```
    printf("Enter player name: ");
```

```
    scanf("%s", search);
```

```
    searchPlayer(players, scores, count, search);
```

```
    break;
```

```
case 4:
```

```
    return 0;
```

```
default:
```

```
    printf("Invalid Choice\n");
```

```
}
```

```
}
```

```
}
```

```
void playGame(int *score)
```

```
{
```

```
    int number = 50;
```

```
    int guess;
```

```
    int attempts = 5;
```

```
    *score = 50;
```

```
    while(attempts > 0)
```

```
    {
```

```
        printf("Enter guess: ");
```

```
        scanf("%d",&guess);
```

```
        if(guess == number)
```

```
        {
```

```
            printf("Correct!\n");
```

```
            return;
```



```

    }

    else if(guess < number)
        printf("Too Low\n");
    else
        printf("Too High\n");

    attempts--;
    *score -= 10;
}

```

```

*score = 0;
printf("Game Over\n");
}

```

These first part is done by the first member Salma

Member 2:

```
#include <stdio.h>
```

```
void displayPlayers(char names[][30], int scores[], int count)
```

```

{
    int i;

    printf("\nPlayer List\n");

    for(i=0; i<count; i++)
    {
        printf("%s %d\n", names[i], scores[i]);
    }
}

```

These second part is done by the member Fariha

Member 3:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void searchPlayer(char names[][30], int scores[], int count, char
search[])
```

```

{
    int i;

```

```
for(i=0; i<count; i++)  
{  
    if(strcmp(names[i], search)==0)  
    {  
        printf("Player Found\n");  
        printf("Name: %s\n", names[i]);  
        printf("Score: %d\n", scores[i]);  
        return;  
    }  
}  
  
printf("Player Not Found\n");  
}
```

These third part is done by the member Shimu